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Enhancing Geological Workflows with AI Assisted Applications

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Abstract

Objectives/Scope: This study aims to develop and validate a fully automated AI-based thin section interpretation model to support reservoir characterization in the SE Abu Dhabi reservoirs. By targeting key carbonate formations - Shuaiba, Kharaib, Lekhwair and Habshan - the objective is to reduce manual interpretation time while enhancing the spatial extrapolation of rock properties. These AI models are designed to assist in field development workflows by providing consistent, scalable insights from petrographic data, ultimately improving the resolution and efficiency of reservoir modeling.

Method: A method for AI-assisted thin section interpretation was developed using advanced deep learning segmentation models and object detection algorithms, all reinforced with geological rules. This solution combines classical regression models and deep learning techniques to provide geologists with a detailed set of reservoir properties derived from thin-section images. The approach includes training supervised convolutional neural networks, establishing consistent labeling procedures, and removing image artifacts from both input and output. Additionally, continuous communication with subject matter experts ensures that predictions remain geologically realistic and help supplement limited training data. In this study, selected thin sections collected from the Shuaiba, Kharaib, Lekhwair and Habshan Formations in an oil field in the SE of Abu Dhabi were analyzed through both standard petrographic observations and the proposed AI-assisted workflow to be able to compare and contrast the obtained results.

Result: The used tool demonstrated an overall strong degree of accuracy when detecting and identifying grains, pore types and abundance, cement features, MICP estimations, depositional environments and textures. Its main limitations appeared to be related to instances of high facies complexity and heterogeneity, or to limited training datasets, which resulted in lower accuracies. However, the tool also provided relatively accurate diagenesis and permeability predictions, which further assisted in achieving a reliable categorization of the analyzed samples into the existing rock typing schemes available for the studied formations. The main outcome of this integration of domain knowledge and data science was an improvement in key evaluation metrics, such as intersection-over-union (IoU), precision, recall, and R^2 . In addition to these performance gains, the approach ensured that fundamental geological principles were maintained. The algorithm constrained petrographic object detection using biostatistical population

criteria, preventing non-natural combinations of nested framework grains and ensuring geologically realistic predictions.

Novel/Additive Information: The application of the proposed AI-assisted petrographic interpretation workflow demonstrated a high potential for streamlining and enhancing the routine analysis of large thin section databases, which often lie underutilized in storage. These enhancements can also be subsequently implemented and deployed at a wider company and even inter-company scale, allowing specialists to conduct and coordinate their geological work through conventional web-browser applications.

Keywords: AI, thin section, petrography, reservoir model

Introduction

Tectonic movements and shifts in sea level during the Early Cretaceous have influenced the stratigraphy of Abu Dhabi, shaping both how sediments were deposited and later altered. A deep understanding of the different rock types (facies), their diagenetic development, and depositional environments is essential for successful hydrocarbon exploration since these formations serve as key reservoirs. By using thin-section analysis and conducting reservoir studies at the microscale, geologists can better identify and interpret important features within these rocks.

In recent years, artificial intelligence (AI) solutions have seen widespread adoption in both the oil and gas industry and geological sciences. Data science algorithm models now facilitate rapid analysis and extraction of insights from geological, geophysical, and production datasets, demonstrating robust capabilities across a range of data types and applications. For example, prior research indicates that such methodologies are highly effective for tasks including seismic fault identification (Egorov, 2019), well log interpretation and localization of unrecognized oil-bearing intervals (Belozarov, 2018), core image analysis for reservoir heterogeneity assessment (Egorov, 2019), and optimization of hydraulic fracture design (Grigorev, 2020), among others. These studies not only illustrate how machine learning can streamline existing business processes but also underscore its capacity to generate new insights from legacy data, thereby reducing geological uncertainties and mitigating field development risks. Nevertheless, despite AI's demonstrated benefits for routine geological activities, there remains limited research focused on the application of such models to thin section analysis (e.g., Bukharev, 2018; Leseur, 2022), highlighting the need for further study and evaluation.

The primary objective of this work is to advance the application of cutting-edge AI methods—specifically, supervised machine learning and computer vision—to optimize microscale reservoir characterization from carbonate rock thin sections in the Lower Cretaceous of Abu Dhabi's oil fields. This investigation encompasses not only the finalized AI models, including their architectures, datasets, training workflows, and validation strategies, but also addresses the challenges encountered throughout the research and development process. The system's reliability is substantiated by both qualitative and quantitative performance metrics.

This study systematically assesses the strengths and limitations of AI-driven approaches compared to conventional petrographic techniques, focusing on the strategies employed to address prevailing challenges. It provides an in-depth comparison, elucidating both the advantages and constraints of incorporating AI into geological research. Additionally, it delivers a comprehensive evaluation of AI's transformative potential for geological workflows, particularly regarding the complexities of carbonate rock formations within Abu Dhabi's reservoirs and their relevance for reservoir characterization. To ensure objective model evaluation, a blind dataset was established to cross-validate performance against predefined parameters.

Introduction to the AI application

An advanced AI-based tool was designed to revolutionize thin section analysis for subsurface reservoir characterization. Built to assist geoscientists and reservoir engineers, it leverages machine learning and computer vision technologies to automate the analysis of geological thin section images. This innovation enables more efficient, accurate, and scalable interpretation of critical rock properties, significantly improving the decision-making process in reservoir exploration, development, and production.

The core functionality of the application is based on its ability to process large volumes of thin-section images and extract essential geological features such as rock types, fossil content, estimated permeability and MICP, diagenetic elements, and interpreted depositional environment. By analyzing these features, the application aids in the identification of key reservoir properties that influence fluid flow, porosity, pore types and overall reservoir behavior. This data-driven approach provides geoscientists with a comprehensive view of subsurface formations, enhancing the precision of reservoir models and predictions.

Through its intelligent automation and sophisticated machine learning algorithms, the tool offers a transformative solution for subsurface analysis. It accelerates the interpretation process, reduces human error, and enhances data consistency, allowing teams to focus on high-level geological interpretation and decision-making. When applied to conventional carbonate reservoirs, the application represents the future of reservoir characterization, optimizing exploration and production strategies in an increasingly complex subsurface environment.

The web-based tool is poised to redefine industry standards by improving the speed, accuracy, and scalability of geological analysis, offering a competitive edge in the fast-evolving field of reservoir management.

Study Goals and Outputs

The study aims to automate and standardize petrographic analysis using advanced AI models tailored for ADNOC reservoirs. Its core objective is to enhance geological interpretation accuracy, reduce manual workload, and ensure consistent, expert-level predictions across diverse reservoir types.

The goal of the tool is to:

- Manage thin section images and their metadata/MICP data.
- Provide dashboards and filtering capabilities on images/metadata/MICP data.
- Automate the thin section images interpretation.

The features of the tool are:

- Import images and metadata/MICP data: The tool offers a UI to upload images in different formats (png, jpeg, etc.) and metadata/MICP data in excel files.
- Interactive dashboard for images and their metadata: The tool allows to apply real time filters and generates statistics on images and related data, and
- Export metadata and predictions: The tool allows to export images metadata and the result of the interpretation as Excel/PDF files.
- User management/ workspace allows to manage multiple users with separated workspaces.
- Reservoirs management: The tool can manage reservoir parameters and use them to enhance automatic interpretation.
- Interpretation tool powered by AI that runs sequentially and displays the results in one window:

AI-Driven Workflow: A multi-stage pipeline that automates texture classification, pore & framework grain detection, permeability & depositional environmental modeling, and SRRT classification.

- Reservoir-Specific Branching: Customized handling of reservoirs where Thamama reservoirs follow a direct SRRT pathway different from non-Thamama reservoirs.
- Texture & Structural Analysis: Initial feature extraction includes framework grain detection, texture classification, and image scale detection, forming the basis for all downstream predictions.
- Pore space & Permeability Modeling: Detection of pore structures and prediction of fluid flow capacity to assess reservoir quality.
- MICP Curve Prediction: The system generates capillary pressure curves to support advanced petrophysical analysis.
- SRRT Classification: Non-Arab reservoirs are processed through Static Reservoir Rock Type classification to support broader geological categorization.
- Integrated Output Generation: All results are merged into a unified interpretation, ensuring coherent and reproducible geological assessments.
- The specialized tool brings scalability, consistency, and speed to geological analysis, enabling data-driven transformation in reservoir characterization.

Methodology and workflow

The AI-powered thin section analysis tool was developed to streamline and enhance geological interpretation workflows. It enables geoscientists to automate labor-intensive tasks, improve accuracy, and accelerate the understanding of subsurface rock properties critical for reservoir characterization and field development planning.

The core capabilities include:

- Automated Image and Metadata Ingestion: Upload and process thin section images alongside metadata and MICP data.

AI-Based Interpretation Models: Sequentially run classification models to identify:

- Texture Classification: Identify the primary texture and the secondary texture if it exists.
- Static Reservoir Rock Type (SRRT): Identify the SRRT of a thin section image.
- Framework grain type abundance classification: Identify the framework grain types.
- Porosity-Permeability algorithms: Estimate the permeability given the measured porosity and the metadata of the target image.
- Pore type abundance classification: Identify the primary and the secondary pore types.
- Mercury injection capillary pressure curves generation: Predict the Mercury Injection Capillary pressure (MICP) and their associated curves from an image and its metadata.

Overview of the AI Solution

The platform is purpose-built to automate thin-section image interpretation and provide consistent, high-resolution geological insights. By combining deep learning, machine learning, and rule-based logic, it delivers an end-to-end solution for petrographic and petrophysical analysis.

The system is designed to dynamically adapt based on reservoir type (see [Figure 1](#)). A core set of models – covering MICP, texture, pore, and permeability – applies to all Thamama reservoirs.

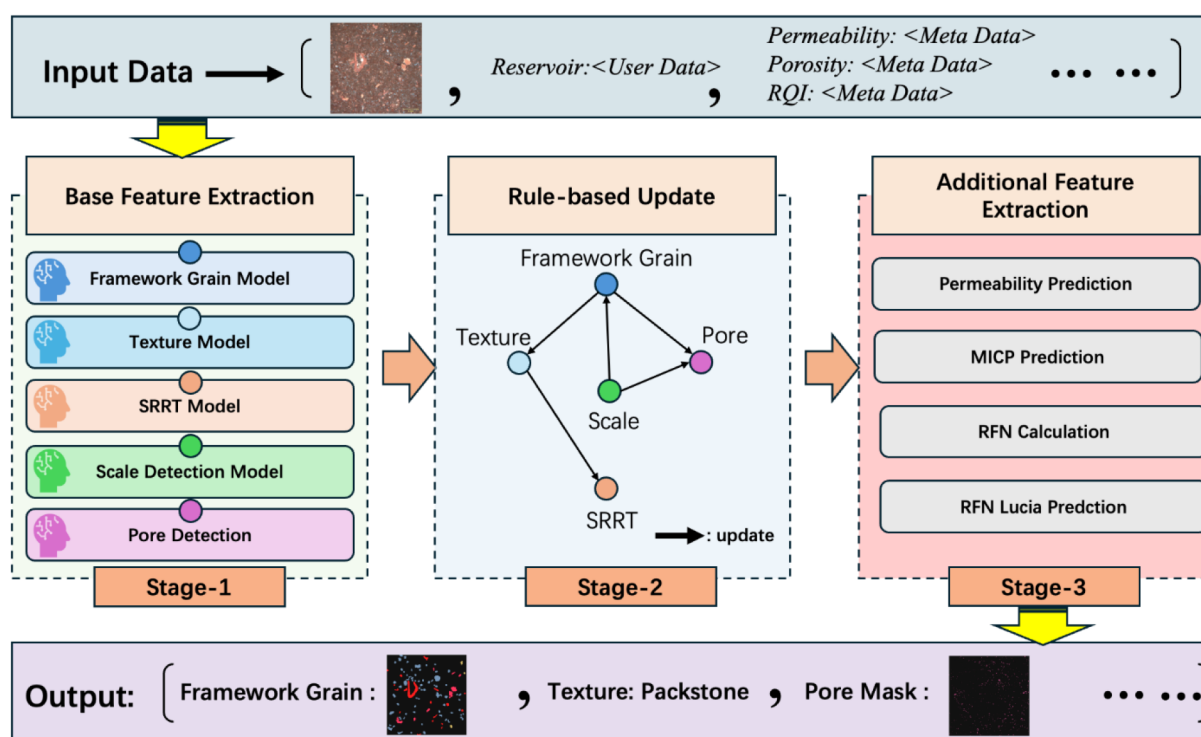


Figure 1—High-level workflow for reservoir characterization and property inference.

Key AI components include:

- **Neural network-based models for texture analysis and SRRT prediction:**

These models use deep convolutional layers to learn multi-scale spatial features from input images. For SRRT prediction, a hybrid approach combines neural outputs with rule-based logic in specific reservoirs, enhancing model interpretability and robustness.

- **Segmentation models for framework grain, cement, and replacement detection:**

These pixel-level classification models use encoder-decoder structures that capture both global context and fine detail. They are optimized for geological imagery and enable accurate identification of rock structures and boundaries.

- **Clustering algorithms and expert-defined rule models for pore structure and porosity estimation:**

Clustering techniques group pore types based on geometry and intensity features, while domain-specific rules guide porosity classification to ensure geological consistency.

- **Ensemble-based regression models for permeability prediction:**

These include gradient boosting regressors trained with cross-validation. Model parameters such as learning rate and number of estimators are optimized per reservoir type, balancing generalization and overfitting.

- **Scale detection and normalization modules:**

These models identify scale bars or known references in the images to normalize spatial measurements, ensuring consistent analysis regardless of input resolution or image source.

The final output integrates all predictions – automatically adjusting the pathway based on reservoir characteristics – to deliver a coherent, expert-level interpretation with minimal manual intervention. By structuring the solution around geological context, the application ensures both scalability and subsurface accuracy across diverse reservoir conditions.

Results and discussion

Figure 1 illustrates the overall architecture of our automated thin section interpretation pipeline, which is structured into three sequential stages to ensure both robustness and geological coherence. Stage-1 (Base Feature Extraction) leverages a suite of specialized deep learning models or machine learning algorithms to extract primary attributes from thin-section images, including framework grain segmentation, texture classification, SRRT estimation, scale detection, and pore structure identification. These outputs serve as foundational descriptors of the input rock fabric. Stage-2 (Rule-based Update) introduces a domain-informed graph-based mechanism that encodes geological interdependencies among features—such as texture, pore morphology, grain framework, and SRRT classification—to perform iterative consistency refinements. This stage ensures that outputs align with expert geological knowledge by enforcing relational constraints and contextual feedback across features. Stage-3 (Additional Feature Extraction) builds upon the updated representations to derive higher-level reservoir properties, including permeability prediction, MICP curve inference, and RFN (Rock Fabric Number) calculations, alongside both conventional and Lucia-based classifications. The final output consolidates all derived features into a structured report, providing interpretable and geologically grounded predictions suitable for integration into digital core analysis and reservoir modeling workflows.

This section summarizes the performance of core models developed for the platform, covering key geological and petrophysical tasks such as grain segmentation and texture classification, permeability estimation, and diagenetic feature detection. The results demonstrate consistent improvements across formations and model versions, with notable gains in both segmentation accuracy and predictive reliability. These outcomes validate the robustness and scalability of the models in supporting carbonate reservoir characterization.

Framework Grain Model

The proposed *Framework Grain Model* exhibits strong generalization capability across multiple carbonate reservoirs. It achieves the highest mean Intersection-over-Union (mIoU) on **Habshan** (0.54) and **other reservoirs** (0.58). Recall values remain above 0.80 for the majority of major reservoirs, underscoring the model's robustness in detecting framework grain features. However, comparatively lower performance on **Lekhwair** suggests that further refinement may be needed to better handle the more heterogeneous or diagenetically complex facies types of this reservoir.

Table 1—Performance of the Framework Grain Model across reservoirs.

Reservoir	mIoU	Precision	Recall
Habshan	0.54	0.85	0.82
Lekhwair	0.31	0.68	0.50
Kharaib-1 Kharaib-2 Shuaiba-1	0.58	0.93	0.90

Texture Model

The proposed *Texture Model* demonstrates consistently strong performance across multiple carbonate reservoirs. On the **Common Texture** group—which includes **Kharaib-1**, **Kharaib-2**, **Shuaiba-1**, **Lekhwair**, and **Habshan**—the model achieves an F1 Score of **0.92**, with high **precision (0.93)** and **recall (0.91)**, indicating accurate and reliable detection of dominant texture types.

Table 2—Performance of the Texture Model across reservoirs.

Reservoir	F1 Score	Precision	Recall
Kharaib-1 Kharaib-2 Shuaiba-1 Lekhwair Habshan	0.92	0.93	0.91

Permeability Model

The *Permeability Model* demonstrates strong predictive capability in several well-characterized carbonate formations. Notably, it achieves high coefficient of determination (R^2) scores in **Kharaib-1** (0.776), **Shuaiba-1** (0.655), and **Kharaib-2** (0.559), indicating reliable regression performance where permeability trends are well-constrained. **Moderate performance** is observed in **Habshan** (0.512) while the results in **Lekhwair** (0.750) reflect potential challenges stemming from greater facies heterogeneity or limited training coverage. These findings underscore the importance of incorporating additional data or geological priors to enhance model robustness in more complex reservoirs.

Table 3—Performance of the Permeability Model across reservoirs.

Reservoir	R^2
Kharaib-1	0.776
Kharaib-2	0.559
Shuaiba-1	0.655
Habshan	0.512
Lekhwair	0.750

Cement Model

The *Cement Model* demonstrates strong and consistent performance in segmenting cementation features across a range of carbonate reservoir types. It achieves a **mean Intersection-over-Union (IoU) of 0.51** and a high **F1 Score of 0.93**, indicating both accurate boundary delineation and overall classification effectiveness. The model also attains high **precision (0.95)** and **recall (0.93)**, suggesting reliable detection with few false positives or missed regions. This balance between precision and recall is particularly important in geological applications, where over- or under-segmentation of diagenetic features can significantly affect subsequent interpretation and property modeling. The results confirm the model's robustness in capturing cementation textures, including pervasive and patchy cements, and its applicability to diverse depositional and diagenetic settings. Overall, the Cement Model offers a reliable foundation for automated or semi-automated diagenetic analysis in complex carbonate systems.

Table 4—Performance of the Cement Model across reservoirs.

Metric	Score
mIoU	0.5097
F1 Score	0.9336
Precision	0.9489
Recall	0.9321

SRRT Model

The updated *SRRT Model (Version 2)* demonstrates substantial performance gains across all porosity classification ranges. Compared to its predecessor, the new model achieves an improved **overall accuracy of 82.3%**, reflecting enhanced predictive capability in both low- and high-end porosity scenarios. Notably, the model delivers significant improvement in the **Low SRRT** range, attaining an accuracy of **87.2%**, and in the **High SRRT** range with **78.9%** accuracy, both representing meaningful advancements over previous results. Meanwhile, performance in the **Medium SRRT** range is effectively maintained at a strong **80.8%**, indicating that the model generalizes well across the full SRRT spectrum. These results confirm the effectiveness of the updated architecture in capturing the variability of porosity-related features and enhancing classification robustness across diverse carbonate facies.

Table 5—Performance of the SRRT Model across reservoirs (V1 & V2).

Model Version	Low	Medium	High	Avg
V1	82.7%	80.8%	71.1%	78.2%
V2	87.2%	80.8%	78.9%	82.3%

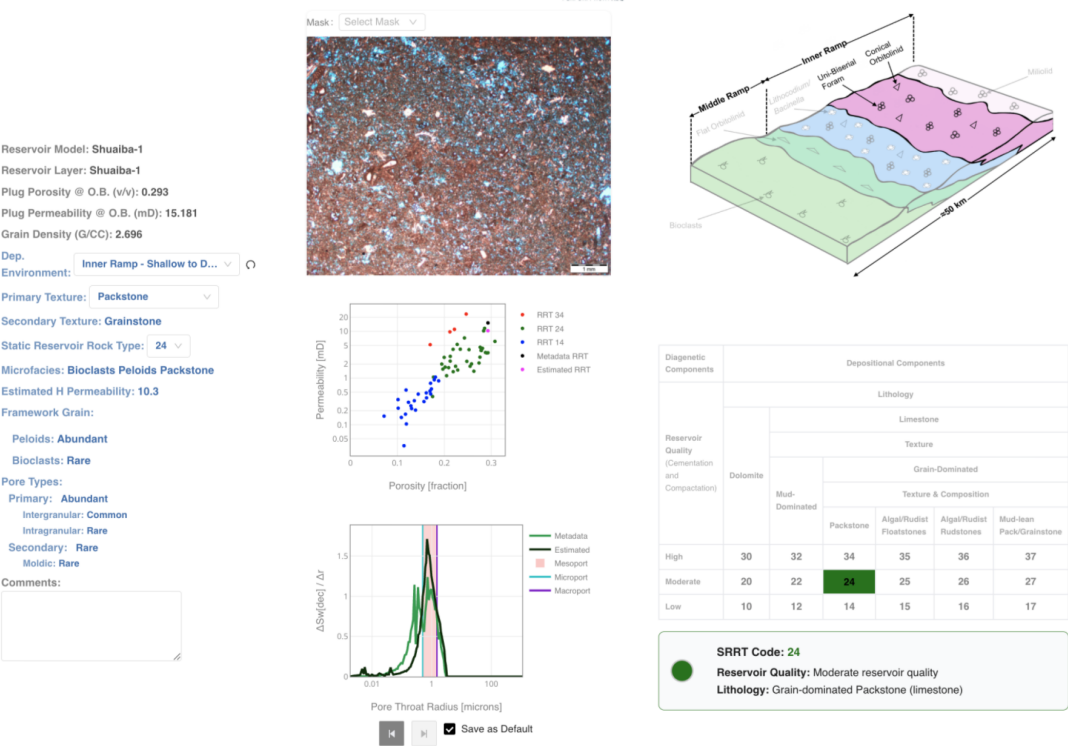
Table 6—Reservoir-wise Accuracy (V2).

Reservoir	Low	Medium	High	Avg
Kharaib-1	81.0%	75.0%	50.0%	68.6%
Kharaib-2	100.0%	96.8%	92.3%	96.3%
Shuaiba-1	100.0%	100.0%	-	100%
Lekhwair	71.4%	-	-	71.4%
Habshan	87.1%	51.8%	50.0%	62.9%

MICP Model

The *MICP prediction algorithm* is a rule-based framework grounded in a **K-Nearest Neighbors (KNN)** retrieval strategy. Rather than learning direct mappings from input features to output labels, the method operates by identifying and retrieving the most geologically similar MICP curves from a curated reference database, based on a predefined set of matching criteria. Due to its non-parametric and heuristic nature, the model does not conform to conventional validation protocols typically used in supervised learning. Instead, its performance is evaluated **qualitatively** through expert-driven geological interpretation, focusing on the geological plausibility and contextual consistency of the retrieved curves. This approach emphasizes interpretability and domain alignment, making it particularly suited for scenarios where ground-truth labels are sparse or ambiguous.

This table presents the model performance metrics (Precision, Recall, F1 Score, and R^2) across three datasets: Habshan, Lekhwair, and Shuiaba-1. Each dataset includes evaluation results for two classification tasks (Texture and SRRT) and one regression task (Permeability). The number of total samples used for each task is listed in parentheses.



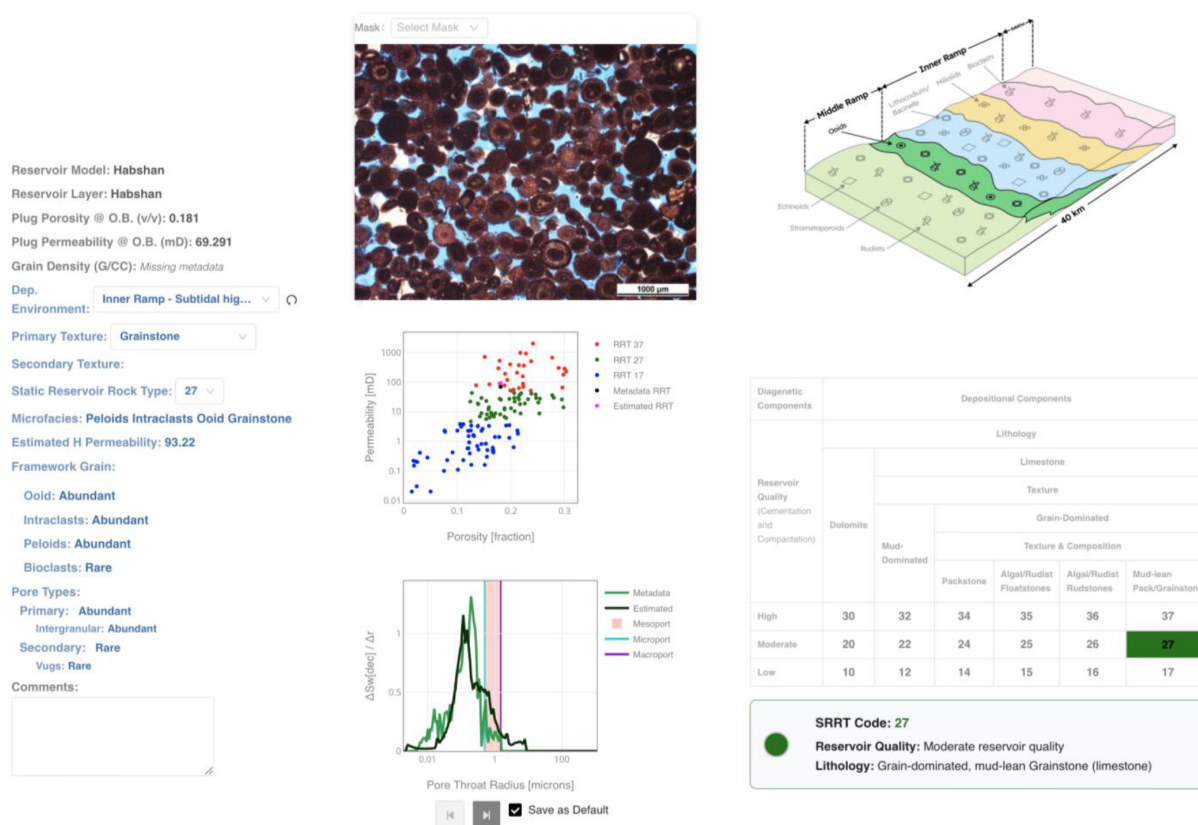


Figure 4—Habshan reservoir model.

Conclusion

This study presents a fully integrated AI framework for automated thin section interpretation, offering a transformative approach to carbonate reservoir characterization in Abu Dhabi's Lower Cretaceous formations. By combining deep convolutional neural networks, regression algorithms, and geologically grounded rule-based logic, the system accurately predicts a wide range of petrographic and petrophysical properties—including texture, framework grains, pore types, diagenesis, permeability, and SRRT classifications—directly from high-resolution imagery. Validated across multiple reservoirs with varied facies complexity, the models demonstrate strong generalization, precision, and scalability. The platform not only accelerates traditionally manual and subjective workflows but also enables the systematic digitization and interpretation of legacy petrographic archives, thus unlocking underutilized data resources. By embedding geological priors and expert constraints within a data-driven architecture, the approach ensures interpretability, reproducibility, and domain alignment. These findings highlight the growing maturity of AI applications in geoscience and lay the foundation for a new class of intelligent, high-resolution tools that can enhance reservoir modeling, reduce uncertainty, and support more informed field development decisions.

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Appendices

mIoU

Definition: Mean Intersection over Union (mIoU) is a widely used evaluation metric for semantic segmentation tasks. It measures the average overlap between the predicted segmentation and the ground truth across all classes.

Formula: For a given class c , the Intersection over Union (IoU) is defined as:

$$IoU_c = \frac{TP_c}{TP_c + FP_c + FN_c}$$

Where:

- TP_c : True Positives for class c (correctly predicted pixels of class c)
- FP_c : False Positives for class c (pixels incorrectly predicted as class c)
- FN_c : False Negatives for class c (pixels of class c missed by the model)

Then, the mean IoU (mIoU) across all C classes is:

$$mIoU = \frac{1}{C} \sum_{c=1}^C IoU_c$$

Intuition: mIoU penalizes both over-segmentation (false positives) and under-segmentation (false negatives). A perfect segmentation yields an IoU of 1 for every class. The metric provides a balanced evaluation, especially in multi-class scenarios with class imbalance.

F1 Score

Definition: The F1 Score is a harmonic mean of Precision and Recall. It provides a single metric that balances the trade-off between false positives and false negatives and is especially useful in classification and segmentation tasks where class imbalance is a concern.

Formula: For a specific class c , the F1 Score is calculated as:

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

Where:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

If evaluating multiple classes, the Mean F1 Score is usually computed by averaging the F1 scores across all C classes:

$$\text{MeanF1} = \sum_{i=1}^C \frac{n_i}{N} F1_i$$

where N is the total number of samples across all classes, n_i is the number of true samples in class i , C is the total number of classes and $F1_i$ means the F1-score for class i .

Intuition: The F1 Score penalizes extreme imbalances between precision and recall. It is particularly useful when you care equally about not missing relevant results (high recall) and not including too many irrelevant ones (high precision).

R²

Definition: The R^2 score measures how well the predicted values from a regression model approximate the actual ground truth values. It quantifies the proportion of the variance in the dependent variable that is predictable from the independent variables.

Formula:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

Where:

y_i : True value for sample i

$\hat{y}_i$: Predicted value for sample i

$\bar{y}$: Mean of all true values

N : Total number of samples

Intuition: The R^2 score compares the performance of your model against a baseline model that always predicts the mean value. It answers the question: "How much better is my model than simply predicting the average?"

$R^2 = 1$: Perfect prediction

$R^2 = 0$: Model does no better than predicting the mean

$R^2 < 0$: Model performs worse than the mean predictor